



Course Title: Programming Fundamentals (Lab)

Section: BSCS F23 B11

Lab Instructor: Syed Ali Zamin	Semester: Spring 2024
Programming language: C++	Marks: 20
Exam: Mid-Term (lab)	Time: 110 Minutes
Date:	Program: BSCS
Instructions: Answer all questions in context of class discussions. State all your assumptions clearly. This is a Closed Book & Notes Exam, No Cheat Sheet is allowed. Use of Electronic Gadgets other than Calculators (Laptops, Mobiles, SmartWatches etc.) is not allowed. Exchange of calculators is not allowed.	

Name: _____

Roll#: _____

Question 1: 04 Marks

You are given two files. One, named “words.txt”, contains a list of words and the other, named “antonyms.txt”, contains their corresponding antonyms (a word opposite in meaning to another). You have to read a word from a file named “text.txt” and its antonym from file “antonyms.txt” and store in another file named “updatedtext.txt”

word.txt

```
happy
good
cold
strong
pleasant
```

antonyms.txt

```
sad
bad
hot
weak
nasty
```

updatedtext.txt

```
Happy
Sad
Good
Bad
Cold
Hot
Strong
Weak
Pleasant
nasty
```

Question 2: 04 Marks

Write a function to read from file named “data.txt” that contains 15 digits of integers. You have to create another file named “no_duplication.txt” and store in it all the digits without duplication.

```
12 13 15 124 32 12 124 13 13
19 23 15 17 3 13
```

“data.txt”

Output4:

```
12 13 15 124 32 19 23 17
```

Question 3: 04 Marks

In your group, you are five friends(A,B,C,D,E,F) studying five same subjects. You have to design a program to get numbers related to each subject through input, for all group members. **Input should be in range [0 to 100]**. Subjects are **DLD, PF, PS, ENGII and MVC**.

- Find average for every subject and then average will be return from the function.
- Find percentage for every friend.
- You have to find the best performing subject from all.
- You have to find the best performing friend of the group.

Question 4: 04 Marks

Online Shopping Cart Management System: Problem: Design a system for managing an online shopping cart. Implement functions for adding items to the cart (maximum 5 items) and calculating the total price. Utilize divide and conquer strategy by breaking down the tasks into smaller, reusable functions. Use value returning functions to calculate the total price and void functions to add items to the cart. Employ function prototypes to declare the functions before defining them. Use formal parameters to pass item details (such as name, price, quantity) to the functions. Utilize reference parameters for updating the cart contents. Also apply 17% GST on each product purchased

Input:

- Commands for adding

Output:

- Updated shopping cart contents and total price after each operation.

Output should look like this receipt:

Welcome to UCP Super Store

Item 1: Notebook Quantity:1 Price: 10 $10*1=10$ price_after_gst: $10+10*.17=X$

Item 2: Pen Quantity:2 Price: 12 $12*2=24$ price_after_gst: $24+24*.17=Y$

Total items: ttl qty: 3 total price: 34 Ttl price after gst: $X+Y$

Question 5: 04 Marks

- **Library Book Management System:** Develop a library book management system using pointers. Suppose there are 5 books [A, B,C,D,E] and their availability status['Y','N','Y','Y','N']. Utilize pointers to efficiently handle book records. Use Case:
 - Update book name: "A" to F
 - Update availability status of "Y" to "N"
 - Search for book: if 'A' is available or not

Hint: use functions/pointers with array arrays

***** Good Luck *****